Enrollment No._____

GANPAT UNIVERSITY

DD(BCA)-(MCA) SEM-V EXAMINATION (CBCS) (NEW)

U55A1WPF: WEB PROGRAMMING FRAMEWORK

Nov - Dec 2022

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

1. Answer the following: (ANY SIX) (30)

A. Explain Magento Architecture in detail.

B. Explain Attribute set with example.

C. List out shipping methods. Explain any one in detail.

D. Write down and explain steps for checkout process.

E. How to use catalog price rules ? Explain it with suitable example

F. What is quick search ? Explain in brief

G. Explain concept of Customer Group in detail.

H. Explain how to create order from admin side with all steps.

2. Answer the following: (ANY TWO) (07)

A. Explain Cross Sell and Up Sell in detail.

B. Explain how to create categories in detail.

C. List out types of Products. Explain any two in detail.

3. Answer the following: (ANY TWO) (07)

A. List out types of Payment Methods. Explain any one in detail.

B. Explain Localization Settings in Magento in detail.

C. Explain Checkout Process in detail.

P.T.O.

4. **Answer the following: (ANY TWO)** **(08)**

A. Write steps to create poll in Magento.

B. Write a short note on Newsletter.

C. Explain Customer Personalization in brief

5. **Answer the following: (ANY TWO)** **(08)**

A. Write a short note on Types of Customers in Magento.

B. Explain update order in detail.

C. Which Information are included to Configuring Customer Accounts

END OF PAPER

Seat No._____

GANPAT UNIVERSITY

DD(BCA-MCA) SEM-V EXAMINATION (CBCS)

U55A2IML INTRODUCTION TO MACHINE LEARNING

Nov - Dec 2022

[Time: 3 Hours] [Total Marks: 60]

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. Answer the following: (Any 5) [30]

1. Explain open() with syntax and use.
2. Write a short note on metacharacters.
3. What is Data Frame? Disucss various way to create Data Frame.
4. What is data visualization? Explain the plotting libraries.
5. Discuss different categories of Machine Learning.
6. Draw and explain architecture of ANN in detail.
7. What is node in KNIME Tool? Discuss the execution state of node.

SECTION-II

2. Answer the following: (Any 2) [12]

1. Discuss tell() with syntax and example.
2. Explain any two methods of re module.
3. What is file? Explain types of file in detail.

3. Answer the following: (Any 2) [12]

1. Which are the functions are used to check the missing values? Explain in detail.
2. List out types of plots. Which graph is used to compare data among different categories explain with example.
3. Discuss various applications of AI.

4. Answer the following: (Any 1) [06]

1. Write down the difference between AI and ML.
2. Discuss any four category of node repository.

END OF PAPER

GANPAT UNIVERSITY
DD (BCA-MCA)SEM-V EXAMINATION (CBCS)
U55A3STQ: Software Testing& Quality Assurance

Nov - Dec 2022

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. **Figures to the right indicate full marks**
2. **Be precise and to the point in your answer**

Q.1	**Answer Following (Any 6)**		(30)
1.1	**Answer the following: True or False (Any 5).Also justify your answer**		
	a) Automated Testing is best suited for Usability testing		
	b) In white box testing, you do not need to execute the program.		
	c) Testing team cannot change the status of the defect to 'Fixed'		
	d) If there are good programmers, the newly developed software may not have any defect		
	e) All software applications need not be tested with very high and equal importance		
	f) Alpha Testing is done at Developer's site.		
1.2	**Match Following (any 5)**		
	1) Aesthetics 2) Testing With Execution 3) Stub 4) SQL Injection 5) Desk Checking 6) Selenium	a) Bottom-up Integration Testing b) Security Testing c) Static d) Dynamic e) Top-Down Integration Testing f) Performance Testing g) Usability / User Interface Testing h) Test Automation	
1.2	Draw V Model diagram and briefly explain what is indicated by arrow keys between left side boxes and right side boxes.		
1.3	Briefly explain cost of quality with types.		
1.4	Briefly explain condition coverage giving example		
1.5	Briefly explain functional and non-functional testing meaning and types		
1.7	Briefly explain Security testing with at least 3 possible attacks.		

1.8	**Fill in the blanks : (Any 5)**	
	i. Boundary Value Analysis is _____technique than Equivalence partition for continuous range of Values. Options (a) Worse (b) Similar (c) Better ii. Standard violation can be found using _____ Testing Options (a) Static (b) White Box (c) Black-box (d) Unit iii. Are we building the product right means _____ Options (a) Testing (b) Validation (c) Verification (d) Execution iv. Initialization, Loops, Control flows etc are tested in _____ testing phase Options: A) Dynamic B) Unit C) Black-Box D) Integration v. A metric that determines % of test cases executed from the total number of test cases prepared is known as _____ Options:(a) Test Design Coverage (b) Test Execution Coverage (c) Automation Coverage (d) Test Case Creation Productivity	
	vi. Testing team will change the status of the defect as _____ , if they find the defect is actually not fixed or reoccurred even if development team mentioned that it is fixed. Options: (a) Closed (b) Cancel (c) Deferred (d) Reopen	
	vii. Splitting complex requirements into smaller components to understand the system is known as _____ Options (a) Requirement Analysis (b) Ambiguity Review (c) Test Design (d)Requirement Itemization	
Q.2	**Answer the following: (Any 1)**	
2.1	Explain Negative testing with examples	**(4)**
2.2	What are the typical challenges in SW testing? (Why exhaustive testing is not possible)?	
Q.3	**Answer the following: (Any 2)**	
3.1	Explain Partial / full Formal Review process in brief (meaning, types, details)	**(10)**
3.2	Prepare decision table for the following requirement Consider reservation function for one event that gives 10% discount to all but person with gold card membership gets additional 10% and silver card membership gets 5% additional. However if the members have not paid membership renewal fees fully then only 3% discount is given	
3.3	Briefly explain State Transition Technique.	

Q.4	**Answer the following: (Any 2)**	(8)
4.1	Briefly Explain Top-down integration technique.	
4.2	Briefly explain User Interface testing covering why it is important, which aspects are considered (any 4) with example	
4.3	Explain Compatibility Testing	
Q.5	**Answer the following: (Any 1)**	(5)
5.1	In a simple examination result processing system, there is a simple input form that takes marks of 2 internal tests and displays % of marks obtained by the student. Maximum marks for each test is 30. Write at least 5 test cases for the given requirement in Test case format (Test Case ID, Precondition, Test Condition, Test Steps, Input/test data, Expected Result).	
5.2	Briefly explain any 4 steps involved in test execution process for system testing.	
Q.6	**Answer the following: (Any 1)**	(3)
6.1	Provide meaning and benefits of automated testing.	
6.2	Explain Risk based testing	

END OF PAPER

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GANPAT UNIVERSITY

DD(BCA-MCA) SEM-V EXAMINATION

U55A4SAD System Analysis and Design

Time: 3 Hours] Nov - Dec 2022 **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

1. **Answer the following: (ANY SIX)** **30**

A. What is System? Explain the characteristics of System.

B. Write short note on Fact finding techniques.

C. Explain the roles of System Analyst.

D. Explain Managing Quality Assurance in detail.

E. Write a short note on Input design.

F. Explain Post implementation Review in brief.

G. Explain all symbols used in DFD in detail.

H. Which are System Concepts essential for any system building?

2 **Answer the following: (ANY ONE)** **5**

A. Explain SDLC in Detail.

B. Explain all types of users in brief.

3 **Answer the following:(ANY ONE)** **5**

A. Draw a Decision Table with example.

B. Differentiate: Physical DFD v/s Logical DFD.

4 **Answer the following: (ANY ONE)** **6**

A. Discuss the Code design in brief.

B. Explain the principle of Form design.

5 **Answer the following: (ANY ONE)** **7**

A. Write a short note on Conversion Methods.

B. Which are testing techniques available? Explain in detail.

6 **Answer the following: (ANY ONE)** **7**

A. Originate a Data flow Diagram of Library Management System

B. Originate a Data flow Diagram for Inventory Management System

END OF PAPER

Enrollment No._____

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DD (BCA)-(MCA) – V EXAMINATION

U55A6DM: DISCRETE MATHEMATICS

Nov - Dec 2022

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

1. **Answer the following: (Any Six)** **(30)**

1) If A={1,2,3,4,5}
B ={X/x^2 -7x + 10 = 0}
C={x/x^2 - 8x + 15 = 0}
Then verify that A ∪ (B ∩ C) = (A ∪ B) ∩ (A ∪ C) .

2) $If\ A = \begin{bmatrix} 2 & -5 \\ 3 & 1 \end{bmatrix} find\ A^2\ and\ A^3$

3) Compute: (A) 12! / 9! (B) 8! / 4! 3!

4) Write a short note on Set Theory.

5) Compute: **(1)** $\begin{pmatrix} 8 \\ 2 \end{pmatrix}$ **(2)** $\begin{pmatrix} 13 \\ 7 \end{pmatrix}$

6) U= {1,2,3,4,5,6,7,8} A= {1,2, 3, 7}
Find A' and Power (A).

7) Write a short note on Graph.

8) $A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & -2 & 5 \end{bmatrix}$ then find (a) AA^T (b) A^TA

2. **Answer the following: (Any Two)** **(10)**

1) Find the number of mathematics students at a college taking at least one of languages French, German and Russian given the following data:

65 study French 20 study French and German

45 study German 25 study French and Russian

42 study Russian 15 study German and Russian

8 study all three languages.

2) Prove any one De Morgan's laws.

3) U={1,2,3,4,5,6,7,8,9}A={1,2,3,4,5,6} ,B={4,5,6,7,8,9}
Find A ∪ B and A ∩ B.

3. **Answer the following: (Any One)**

1) Suppose a license plate contains two letters followed by three digits with the first digit not zero. How many different license plates can be printed?
Plate number like ex : GJ 1 68

2) How many ways can the letters can be arranged?
(a) BABBY (b) BENZENE

4. **Answer the following: (Any One)** **(06)**

1) Discuss Subgraph and Homeomorphic graph in detail.

2) Explain Bipartite graph in detail.

5. **Answer the following: (Any Two)** **(08)**

1) $A=\begin{bmatrix} 2 & 2 & 5 \\ 4 & -3 & 7 \end{bmatrix}$ and $B=\begin{bmatrix} 5 & -7 & 6 \\ 4 & -5 & 8 \end{bmatrix}$ then find
(A) -2A-3B (B) A^2

2) Define Terms: Square Matrix, Row Matrix, Unit Matrix, Transpose of matrix

3) If $A=\begin{bmatrix} 6 & 4 & -4 \\ 3 & 3 & 7 \end{bmatrix}$ $B=\begin{bmatrix} 2 & 6 & -5 \\ 7 & 4 & 5 \end{bmatrix}$ find (a) A+B (b) 6A

END OF PAPER